Graphic Era
HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

End Term (Even) Semester Regular/Back Examination May 2026

Roll no...22615.14.........

Name of the Program and semester: B.Tech(CSE) VIII
Name of the Course : Mobile Computing
Course Code: TOE 811

Time: 3 hours

Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. **(CO1)** (2X10=20 Marks)

a. Explain in detail the issues and challenges in mobile computing. Discuss how mobility, bandwidth, power and security affect system design.
b. Discuss the architecture of GSM in detail. Explain the functions of MS, BTS, BSC, MSC, HLR, VLR, AuC and EIR.
c. Explain soft handoff and hard handoff. Compare them and explain why CDMA supports soft handoff.

Q2. **(CO2)** (2X10=20 Marks)

a. Give a detailed overview of Wireless LAN (WLAN). Explain the MAC layer issues in WLAN and describe how IEEE 802.11 addresses these issues.
b. Explain the concept of data broadcasting in wireless systems. Discuss different broadcasting techniques and their advantages in mobile computing environments.
c. What is WAP (Wireless Application Protocol)? Explain the need for WAP and describe its overall architecture.

Q3. **(CO3)** (2X10=20 Marks)

a. What is data replication in mobile computing? Explain its importance. Discuss different replication strategies and how they help in improving availability and fault tolerance.
b. Describe the architecture of the CODA file system. Explain its key features such as hoarding, caching, and replication. How does it support mobile users?
c. What are disconnected operations in mobile computing? Discuss their need and advantages. Explain how systems maintain data consistency when reconnection occurs.

Q4. **(CO4)** (2X10=20 Marks)

a. Discuss the different types of mobile agents. Explain their roles and applications in distributed systems.
b. What is fault tolerance in mobile computing? Discuss different types of failures (network failure, device failure, software errors) and their impact on mobile systems.

c. Explain various security mechanisms for mobile agents, including authentication, encryption, digital signatures, and sandboxing. How do these techniques ensure secure execution?

Q5. **(CO5)** (2X10=20 Marks)

a. Explain the concept of Ad Hoc Networks. Discuss their characteristics, advantages, limitations, and differences from traditional wireless networks.
b. Explain the Ad Hoc On-Demand Distance Vector (AODV) routing protocol in detail. How does it differ from DSR and DSDV? Discuss its advantages and limitations.
c. Describe the Global State Routing (GSR) protocol in detail. Explain its working mechanism, advantages and limitations.